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Explainable AI Cloud Security Framework with Automating SIEM Event Management

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***Abstract*—We study how AI-driven security automation can be systematically designed for resource-constrained cloud environments such as small business and medical environments in remote areas. Existing SIEM deployments often rely on loosely coupled integrations and manual triage workflows, lacking formalized closed-loop orchestration under latency and operational constraints.**

We propose a deployment-aware AI security orchestration framework that formalizes the trigger chain (event → triage → action → visualization → refinement) as a latency-aware closed-loop workflow. Unlike conventional engineering integrations, our approach treats operability and inference latency as first-class design objectives. Our novelty lies in the formalization of an operationally constrained closed-loop orchestration and its measurable system-level metrics, rather than the specific vendor tools used

The framework introduces three contributions.

First, a system-level orchestration model that bridges AI inference with cloud deployment through explicit workflow formalization under operational constraints

Second, a deployment-aware model training and selection strategy that structures event attributes into categorical, contextual, and behavioral feature groups and frames model choice as a multi-criteria decision problem balancing predictive accuracy, inference latency, computational overhead, and interpretability. Model outputs are mapped into operational risk tiers via a defined transformation function, enabling direct integration with automated response policies.

Thirdly, a quantitative operational evaluation framework defining end-to-end workflow latency and reduction of manual intervention as measurable system metrics.

The experiments on a large-scale labeled security event dataset demonstrate that the proposed framework reduces manual intervention and improves response efficiency while maintaining high classification precision.

In this work refers both to the repeatable deployment of the cloud security pipeline and to the reproducibility of experimental results within the deployed environment. All components, datasets, and procedures are documented; all experimental artifacts are publicly released to ensure reproducible deployment and transparent validation. It provides a complete Project reproduction repository (including data, attack examples, experiment results, Analysis charts, dashboard demo), an available for local-run dashboard program code, and a steps operation manual for reproduce and verification. (Link: weijunma/AI-Driven-Cloud-Security-Framework-SME-Healthcare at V1.1.0 DOI 10.5281/zenodo.17605067).

Keywords—adaptive security policies, AI-driven security, cloud computing security, explainable AI, machine learning, deep neural network, SIEM automation, Threat detection Security and Privacy, Social and Professional Topics, Automation.

I. INTRODUCTION

Small- and medium-sized enterprises are gradually incorporating cloud computing as the core of their IT infrastructure. Although cloud platforms enable distributed service delivery, they also expand the attack surface and introduce new operational risks. Furthermore, organizations with limited IT resources, such as medical clinics, rehabilitation centers, and nursing homes located in remote areas, lack dedicated cybersecurity teams or simply rely on a IT support model combining professional remote support with non-Tech staff in the branches. This model may lead to a series of problems, including delayed threat detection and inconsistent policy execution. It not only slows down the response to threat incidents but also endangers the continuity of sensitive data and services.

Although advanced tools such as Microsoft Defender for Cloud, Microsoft Sentinel, and Azure Machine Learning can provide robust threat detection, log management capabilities, they face difficulties in joint deployment, especially in the aforementioned environments. These platforms typically require high technical capabilities for integration and orchestration, assuming the user organization has the corresponding resources and capabilities to directly configure complex rules, write scripts, and conduct real-time log analysis. However, these capabilities often do not exist in such IT resources environments. Additionally, some enterprises with limited resources also attempt to solve the problem through static security information and event management (SIEM) systems. Still, SIEM systems rely on predefined rules, which makes them unable to adapt to

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constantly changing threat patterns, resulting in new problems such as high false alarm rates and further delays in response.

Using an AI-driven framework for recommending response measures is a current trend [14, 17]. In particular, recent data sets—such as the GUIDE data set [11], which contains over one million an-notated security incidents—make it possible to build an AI-driven framework for predicting event classification results and for recommending response measures. Unfortunately, most existing methods focus on optimizing individual components, lacking a complete end-to-end, low-code solution that supports non-technical personnel working in resource-constrained environments.

To fill this gap, we propose an adaptive AI-driven cloud security automation framework that has been verified in a resource-constrained remote medical environment. Our framework integrates commercial cloud services into a closed-loop system that handles the entire process from data collection to threat classification and automatic response. It has also been equipped with a locally deployable web interface, empowering non-technical users and providing real-time, interpretable security situation insight analysis. By integrating cloud services, our framework enables non-technical personnel to perform real-time AI-assisted threat detection and automated SIEM event management. The experiments with the data set, which contain over one million labeled events covers a mixture of intrusion attempts, identity-related events, and system-level security alerts commonly observed in enterprise and healthcare cloud environments., using Extreme Gradient Boosting (XGBoost), random forests, and neural networks demonstrate that our framework can effectively reduce human intervention and is highly scalable, helping to enhance security.

The contributions of this work are as follows:

A. *System-level architectural innovation through closed-loop orchestration.*

We propose a closed-loop AI-driven cloud security orchestration framework that integrates event ingestion, risk classification, automated remediation, visualization, and feedback refinement into a unified operational pipeline. Unlike loosely coupled integrations of existing tools, the proposed design formalizes the trigger chain (event → triage → action → visualization → rule/model refinement) as a latency-aware security workflow. Furthermore, the framework explicitly prioritizes non-expert operability through a locally deployable dashboard, structured automation templates, and maintainable low-code workflows, positioning usability as a first-class design objective rather than an auxiliary interface feature.

B. *Deployment-aware model training and feature-engineering strategy.*

We introduce a deployment-aware training methodology tailored to resource-constrained environments. GUIDE event attributes are structured into categorical, contextual, and behavioral feature groups, followed by preprocessing and stratified sampling to preserve class balance. Model outputs are mapped into operational risk tiers using a defined risk-level transformation function, enabling direct integration with automated response policies. Rather than optimizing solely for predictive accuracy, model selection is framed as a multi-criteria decision process considering accuracy, inference latency, computational overhead, and interpretability. This formalizes the trade-offs between XGBoost, Random Forest, and DNN under practical deployment constraints.

C. *Quantifiable operational impact with reproducible validation.*

We define and measure end-to-end security workflow latency from alert generation to automated remediation execution, and quantify reductions in manual intervention steps enabled by the proposed orchestration. Experimental evaluation demonstrates measurable improvements in response time and operational efficiency within a reproducible deployment setting. All components, datasets, and experimental procedures are provided to ensure repeatability and transparent validation.

II. Background

This section describes the limitations of existing solutions and the need for an integrated approach to address cybersecurity in organizations with limited IT resources.

A. *Existing platforms lack full-process automation.*

Microsoft Defender for Cloud offers multi-level threat detection and compliance monitoring, but its AI functions focus on risk scoring and alarm generation. Therefore, it lacks an automated closed loop from detection to response [9, 10]. Microsoft Sentinel, although having strong log aggregation capabilities, relies on static association rules. This reliance leads to a high false positive rate and requires a large number of IT personnel for manual debugging and optimization [13].

B. *Current research addresses only a portion of the issues.*

Calles et al. proposed a serverless security model across multiple clouds, but they failed to achieve end-to-end automation [7]. Kalyva focuses on protecting Azure Machine Learning models, but did not address SIEM integration [14]. In the medical field, a prime example of resource-constrained IT organizations particularly remote clinics, lack of comprehensive solutions is even more common. Algarni and Thayananthan [1] applied explainable artificial intelligence (XAI) to communication security, but their research did not achieve extensive security automation. A review by Ferrag et al. [10] acknowledged the security potential of large language models (LLMs), but studies that can be actually deployed are still scarce.

C. *Some data sets facilitate hierarchical AI research, but shortcomings remain.*

Microsoft's Copilot Guided Response (CGR) system uses a random forests model combined with a large amount of pre-processing. Among them, it also includes high cardinality feature compression, one-hot encoding, and principal component analysis (PCA). [12] The Microsoft Global Incident and Detection Environment (GUIDE) data set [11] contains over 1 million labeled events, but the research based on GUIDE is limited to traditional machine learning. Moreover, its interpretability is limited, which means that users must possess considerable technical proficiency. These issues directly lead to our work, which not only integrates and deploys Microsoft Defender for Cloud, Microsoft Sentinel, and Azure Machine Learning but also uses them in combination with random forests, XGBoost, and deep neural network (DNN) models, providing an

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interpretable end-to-end automated solution for organizations with limited IT resources, including experienced cybersecurity professionals.

III. Framework

Our AI-driven cloud security framework specifically targets small and medium-sized enterprises with limited IT resources, such as small medical or rehabilitation institutions in remote areas. This section describes the architecture and workflow of our system

A. Architecture

We integrate commercial cloud services, specifically Microsoft De-fender for Cloud, Microsoft Sentinel, and Azure Machine Learning, into an end-to-end closed loop and can handle the process from threat detection to intelligent incident response. Integrating these services significantly reduces the reliance of the aforementioned organizations on professional IT resources and dedicated security teams.

- Design Principles. Our framework adheres to the following core principles:
- Functional modularization Each functional layer can operate and be upgraded independently so that it does not affect the stability of other layers.
- Minimize barriers to entry Non-technical personnel should be able to directly operate the system with minimal training.
- Prioritize cost control and efficiency We leverage existing li-censing agreements, such as Microsoft cloud services, to avoid additional infrastructure investments.

We selected the aforementioned cloud services for the following reasons. Microsoft Defender for Cloud is well integrated with the native security control of Azure, which enables unified monitoring of vulnerabilities and compliance. Microsoft Sentinel provides SIEM functionality on a pay-as-you-go basis. This advantage can significantly reduce upfront costs, especially compared to local deployments, such as Splunk. Azure Machine Learning seamlessly connects with Sentinel logs through log analytics. This interoperability enables low-code machine learning (ML) model deployment quickly, and it does not require maintaining an independent computing cluster.

B. System Design.

As shown in Figure 1, our five-layer design integrates cloud security protection tools (Microsoft Defender for Cloud), an alert system (Microsoft Sentinel), and machine learn-ing (Azure), forming a complete closed-loop system. The system not only covers detection, analysis, and automation, but also adds AI and visualization components. Therefore, it can be applied to small and medium-sized enterprises with limited resources and the healthcare sector, and enables practical deployment by non-technical personnel

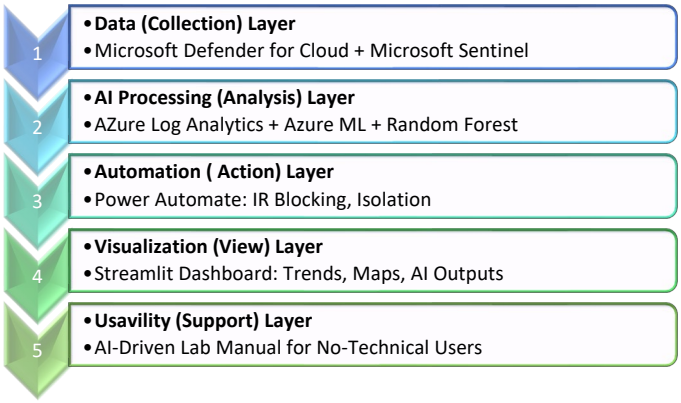


Fig. 1. Overview of the proposed AI-based cloud security framework design.

Data collection: The framework continuously monitors virtual machines, storage, and network components through Microsoft Defender for Cloud. Additionally, alerts and logs are aggregated by Microsoft Sentinel to ensure the standardization of event data for subsequent AI processing

AI processing: Event data is imported into Azure Log Analytics after which classification is performed using a ML model trained on the GUIDE data set. Supporting multiple ML models, such as XGBoost, LightGBM, and DNN, enables future adaptation to the continuous changes in threat pat-terns.

Automation: Based on the AI classification results, Microsoft Power Automate performs security operations such as blocking, isolation, and remediation. Using Power Automate re-duces the maintenance cost of custom scripts and enables non-programmers to update the workflow, thereby meeting the needs of resource-constrained organizations.

Visualization: A Streamlit dashboard displays real-time at-tack trends, geographical distribution, and interpretable AI outputs. This layer is intentionally decoupled from the core automation functions so that interface adjustments will not affect the detection and response capabilities.

Usability By providing operation manuals and step-by-step guidance, non-technical personnel can independently deploy and operate the system

The modular design of this framework supports future integration with LLMs. Additionally, the modular design also helps the framework achieve natural language-based SIEM analysis and strategy optimization. Thus, it ultimately enables compatibility with public, hybrid, or private cloud deployments

C. AI Driven Security Automation Workflow

The workflow is divided into five consecutive steps to minimize the delay from detection to response, while enhancing operational efficiency. For example, a high-risk alert classified by the model can automatically trigger actions such as blocking a suspicious IP via firewall rule updates, isolating a compromised virtual machine by modifying network security group policies, and initiating remediation workflows through predefined automation runbooks:

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Combine threat detection and log aggregation: It use Microsoft Defender for Cloud for threat detection, and then aggregate the findings with Microsoft Sentinel. Although the log structure needs to be structured into a JSON format and the triggering method selects the event-driven data access pipeline approach, the average log access delay time is less than 10 seconds.

1) AI-driven risk classification Events are classified (true) positives, benign positives, and false positives) using the Azure Machine Learning random forests model, and then assigned a risk level. The input includes structured logs of contextual information (such as IP, geographical location, etc.). The output includes log analytics tables that store event records with risk labels. The average execution time for batch processing 500 events is approximately 2.1 seconds.

2) Risk-tier mapping definition To operationalize model outputs within the automation pipeline, we define a risk-tier mapping function. Let P_c denote the predicted probability for class c, and S_c represent the severity weight associated with that class, derived from operational response categories. The operational risk score is defined as $R = max_{\ c}(P_c \cdot S_c)$. Risk tires are then assigned based on predefined thresholds, enabling direct integration with automated response policies.

Let:

$$P_c = predicted\ probability\ for\ class\ c \quad (1)$$

$$S_c = severity\ weight\ assigned\ to\ class\ c \quad (2)$$

Define operational risk score:

$$R = \max_{c} \quad (Pc \times Sc) \quad (3)$$

Then define risk tiers:

High if R> T_h

Medium if T_m < R < T_h

Low otherwise

Severity weights are assigned according to predefined operational response categories aligned with incident criticality levels, ensuring that higher-impact security events contribute proportionally to the operational risk score. Thresholds for tier assignment are determined based on deployment-specific response policies.

Automated response steps Microsoft Power Automate executes rule updates for the firewall and account isolation (or alert escalation) based on the AI classification results. The trigger method for the experimental design is the AI classification output, and the execution is completed through Azure Functions and the Defender API. The results show that the response is initiated within <5 seconds after classification.

Visualized steps and manual supervision The data up-date method invokes the log analytics API every 15 seconds through the Streamlit dashboard. The dashboard displays attack patterns and risk distribution and provides explain-able AI outputs (feature importance, SHAP visualization, etc.). Event analysis and compliance report export is very practical for healthcare institutions.

Empower users through the dashboard and operation manual Enabling non-technical personnel to independently manage the security workflow is critical for organizations with limited IT resources. When necessary, model re-training can cover the automation rules and annotate event records. This feedback loop continuously adapts and optimizes for localized security requirements.

This framework optimizes and integrates the workflow through a hierarchical architecture, resulting in a self-reinforcing, adaptive, and cloud-based security system. In resource-constrained environments, it can significantly enhance the speed of incident response while improving the overall security resilience.

IV. Evaluation

Our evaluation considers three perspectives. First is the accuracy of threat detection. Second is the explainability of the model. Third is an analysis of security incident trends.

A. Experimental Setup

Our experimental environment replicates the core functions of the Microsoft Azure security ecosystem to meet the needs of resource constrained organizations, such as small- and medium-sized enterprises and remote medical institutions. The main services include event monitoring, SIEM-style log aggregation, model training and deployment, low-code automated orchestration, and visual dashboards using Microsoft Defender for Cloud, Microsoft Sentinel, Azure Machine Learning, Microsoft Power Automate, and Streamlet dashboards respectively. We not only verify technical feasibility but also evaluate the usability of the proposed framework in resource constrained scenarios through this environment.

All experiments were completed on workstations equipped with Nvidia Tesla V100 GPUs.

B. Data Sets.

Our core experiment utilizes the Microsoft Global Incident and Detection Environment (GUIDE) data set [12],1 which contains over one million security events labeled by professional analysts. The labels include true positives (TP), benign positives (BP), and false positives (FP). We use stratified sampling to ensure a balanced distribution of the training set and the test set. To further align with the real scenario, we also generated synthetic logs covering various types of intrusions including bruteforce attacks, SQL injection, and credential stuffing. These logs also include contextual fields (e.g., response time and user agent).

C. Synthetic Log Generation and Schema Alignment

To complement the Microsoft GUIDE dataset and better approximate real-world SOC/SIEM scenarios, we generate controlled synthetic security logs that simulate additional attack behaviors and operational contexts.

Synthetic logs are constructed using scenario-driven templates derived from common enterprise and healthcare threat patterns, including brute-force authentication attempts, SQL injection, credential stuffing, lateral movement, and abnormal privilege escalation. Each template defines structured event fields consistent with SIEM log schemas (e.g., timestamp,

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source IP, destination IP, port, user ID, device identifier, geographic location, event type, response latency, and user-agent string).

To ensure compatibility with the model pipeline, synthetic events are mapped to the same categorical, contextual, and behavioral feature groups used in GUIDE-based training. Feature alignment includes:

Categorical features (e.g., event category, authentication outcome)

Contextual features (e.g., IP range, geographic metadata)

Behavioral features (e.g., event frequency within time window, session anomaly indicators)

Class distributions are controlled to simulate varying operational conditions. We generate both balanced and skewed distributions to approximate real-world alert imbalance commonly observed in SOC environments. Synthetic logs are integrated into the deployment pipeline using the same ingestion and processing workflow as GUIDE events, ensuring consistency in feature extraction and risk-tier mapping.

D. Stress Test Under Distribution Shift (Concept Drift)

To evaluate robustness beyond the baseline GUIDE distribution, we conduct a controlled stress test under simulated concept drift conditions.

Specifically, we modify the synthetic log distribution in three ways:

Attack-type shift: Increasing the proportion of identity-related incidents while reducing intrusion-based alerts. Geographic drift: Changing IP-origin distributions to simulate new regions not dominant in training data. Behavioral frequency shift: Altering temporal burst patterns to emulate evolving attack intensity. The model is trained on the original GUIDE dataset (and baseline synthetic augmentation) and evaluated on drifted synthetic distributions without retraining. Drift scenarios are simulated by controlled redistribution of synthetic log parameters as described in Section A.2.

Results indicate that classification performance exhibits moderate degradation under increasing distribution shift, particularly in the combined-drift scenario. However, the closed-loop orchestration framework maintains stable end-to-end workflow latency and preserves a substantial reduction in manual intervention. Notably, operational latency increases by less than 0.3 seconds across drift scenarios, demonstrating that the system-level design and risk-tier mapping remain computationally stable under moderate distribution perturbations. These findings suggest that while model-level accuracy is affected by distribution shift—as expected—the operational robustness of the deployment-aware orchestration architecture remains resilient, supporting practical deployment in evolving threat environments.

The synthetic templates follow common SOC/SIEM field structures and are parameterized to match GUIDE-derived marginal distributions.

TABLE I. PERFORMANCE UNDER DISTRIBUTION SHIFT

Scenario	Precision (%)	Recall (%)	F1-score (%)	End-to-End Latency (s)	Manual Intervention Reduction (%)
Baseline (GUIDE)	89	87.5	88.2	4.8	42.3
Attack-Type Shift	86.7	84.9	85.8	4.9	40.1
Geographic Drift	85.9	83.8	84.8	4.9	39.7
Behavioral Frequency Shift	84.6	82.3	83.4	5	38.5
Combined Drift (Worst Case)	82.8	80.9	81.8	5.1	36.9

The comparative results consolidated in Table IV provide quantitative evidence of the proposed framework's operational resilience under realistic distribution shift conditions. Across all simulated drift scenarios—encompassing attack-type shift, geographic drift, behavioral frequency shift, and their combined worst-case configuration—the framework exhibits a consistent yet moderate degradation in classification metrics, with the F1-score declining from 88.2% under the baseline GUIDE distribution to 81.8% in the combined-drift scenario. Crucially, end-to-end workflow latency remains tightly bounded, increasing by no more than 0.3 seconds across all conditions, while manual intervention reduction remains substantial, ranging from 42.3% at baseline to 36.9% in the worst case. These results directly substantiate the central thesis of this work—namely, that AI-driven security automation can remain operationally robust even in the presence of non-stationary threat distributions. The risk-tier mapping function and the closed-loop orchestration architecture together insulate system-level performance from the model-level accuracy perturbations inherent to concept drift, affirming the practical deployability of the proposed framework in dynamic, resource-constrained healthcare and SME environments [16, 17].

E. Models

Neural Networks. Artificial neural networks (ANN) are a well-known artificial intelligence technique that seeks to mimic the structure and activity of neurons in the human brain. Several disadvantages of neural networks are that they can be both time consuming to train and prone to over-fitting; however, they can theoretically compute arbitrary functions, making them capable of learning non-linear functions. Generally, improvements in ANN performance are achieved through experimentation with different activation functions, loss functions, batch sizes, etc. [15].

XGBoost. eXtreme Gradient Boost, or XGBoost, belongs to the gradient boosting family of techniques [8, 9]. At a high level, Gradient Boosting is the application of an optimization algorithm to a cost function [4]. XGBoost builds multiple sequential learners, with each succeeding learner attempting to correct the error made by the previous learner [8, 9]. The technique consists of multiple decision trees [15].

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Random Forest. Random Forests leverage averaged performance over many tree classifiers (hence, a "forest") to improve the model's overall performance [2]. The values assigned to tree decision nodes are independently sampled from a distribution for each new tree, the purpose of which is to prevent over-fitting [5, 6]. The technique works by fitting a suite of decision trees to the given data, where each of the trees is fitted to a random subset of the data [16]. Each node of each tree then uses a subset of the inputted features to determine whether to expand the tree [16]. The overall prediction is then calculated by taking the average of the predictions from the "forest" of decision trees [16]

Metrics. We include a number of metrics to assess the effectiveness of our system.

Accuracy. Accuracy measures the proportion of correct classifications (both true positives and true negatives) among the total number of cases examined. It provides a straightforward indicator of the model's overall performance in correctly identifying both normal and anomalous traffic.

$$accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (4)$$

where TP, TN, FP, and FN represent the numbers of true positives, true negatives, false positives, and false negatives, respectively. Accuracy can be biased by the majority class (e.g., benign traffic or TN).

ROC-AUC score. The receiver operating characteristic (ROC) curve visualizes the model's performance across all classification thresholds, and the area under the curve (AUC) quantifies this performance into a single metric. The ROC-AUC score is a measure of the model's ability to distinguish between the classes, with a score closer to 1.0 indicating a better model.

F_1 score. The F_1 score is the harmonic mean of precision and recall, offering a balance between the two metrics. Precision quantifies the number of true positive results in relation to all positive results returned by the model:

$$precision = \frac{TP}{TP+FP} \quad (5)$$

Recall, or sensitivity, measures the proportion of actual positives that are correctly identified by the model:

$$recall = \frac{TP}{TP+FN} \quad (6)$$

The F_1 score is particularly useful in scenarios where there are significant class imbalances, as is often the case in anomaly detection tasks.

$$F_1 = 2\left(\frac{precision \times recall}{precision+recall}\right) \quad (7)$$

Precision and recall (and, by extension, the F_1 score) mitigate the base rate fallacy for the effectiveness of a system [3].

F. Results

1) Threat Detection.

The random forests classifier performs exceptionally well on the synthetic data set. As shown in Tables I and II, the majority of the classifications are correct (true positives and true negatives) and the accuracy rate is 92.5%. Other metrics exceed 93%. The results indicate that the recall rate of the attack detection is acceptable with few false negatives. Especially when noise and randomization features are added, the results remain stable.

Model Comparison. We also compared the performance of each model (random forests, XGBoost, and DNN) using the Microsoft GUIDE data set. Figure 2 shows the multi-class ROC curve, and Table III presents accuracy, precision, and recall. Although XGBoost had the best accuracy (89%), it has a high computational cost. Random forests, which has a slightly lower accuracy (81%), is more advantageous when deployed in resource-constrained environments.

From a system deployment perspective, our framework prioritizes low-latency response, operational simplicity, and cost efficiency in resource-constrained healthcare environments. Although XGBoost achieves slightly higher accuracy, Random Forest is selected for deployment due to its lower inference cost, faster response time, and simpler operational maintenance, which are critical in resource-constrained healthcare environments. Therefore, although XGBoost achieves slightly higher accuracy (89%), Random Forest is selected for the operational workflow due to its lower computational overhead and faster inference, making it more suitable for real-time automation.

TABLE II. CONFUSION MATRIX FOR THE RANDOM FORESTS MODEL ON

	Predicted	
	Attack	***Benign***
Attack	28200	1800
Benign	1600	16400

TABLE III. PERFORMANCE OF THE RANDOM FORESTS MODEL

Random Forests	**Random Forests Percentage**			
	Accuracy	***Precision***	***Recall***	***F_1 score***
Percentage (%)	92.5	93	94	93.5

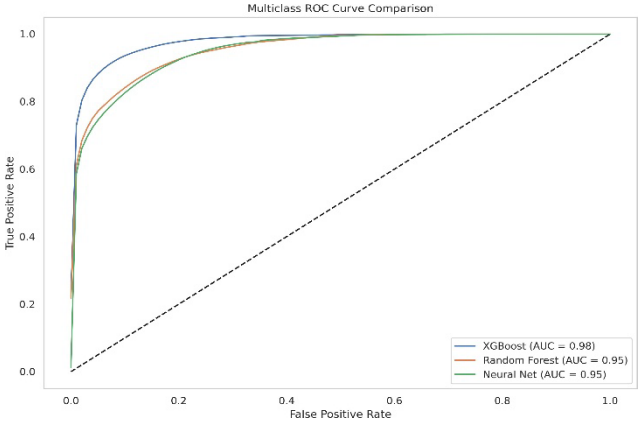


Fig. 2. Multiclass ROC Curve Comparison

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TABLE IV. DETECTION ACCURACY COMPARISON BETWEEN MODELS

Models	Detection Accuracy Percentage		
	Accuracy	*Precision*	*Recall*
XGBoost	89	89	89
Random Forests	81	85	81
DNN	82	83	82

2) Results of Feature and Explainability.

The Gini importance analysis of random forests, as shown in Figure 3, indicates that the features, such as the organization identifier, incident identifier, and detector identifier, play a dominant role in the classification decision. These explanatory results will undoubtedly enhance users' trust in the model and the interpretability of its operation.

3) Analysis of Trends and Categories of Security Incidents.

After analyzing the time distribution and types of incidents, we discovered an interesting phenomenon, as shown in Figure 4, that attacks are more concentrated during working hours. This indicates that real-time defense capabilities are of utmost importance.

The distribution of various attack incidents is shown in Figure 5. Initial access and exfiltration account for the majority of events so these categories should be prioritized as the focus of protection.

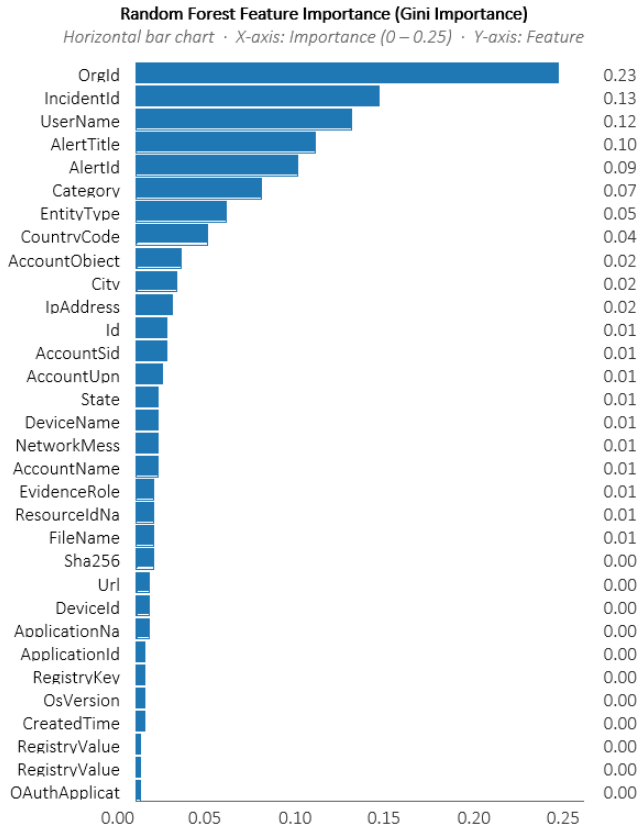


Fig. 3. Feature Importance (Gini) of the random forests model

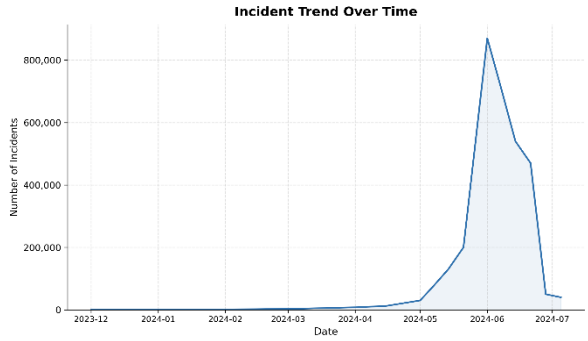


Fig. 4. Distribution of the time trend of security Incidents. Incident Trend Over Time

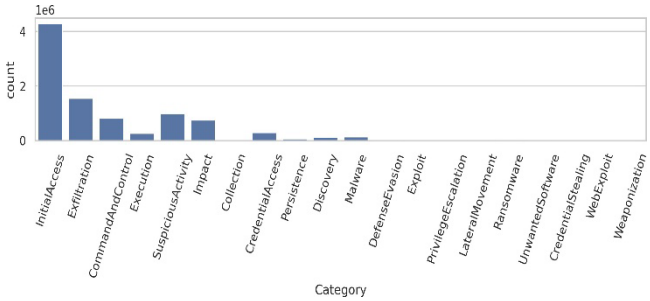


Fig. 5. Distribution of events for each attack category

V. CONCLUSION AND FUTURE WORK

We have described an AI-driven cloud security automation framework specifically for small and medium-sized enterprises and medical institutions with limited technical resources. This framework not only integrates Microsoft Defender for Cloud, Microsoft Sentinel, and Azure Machine Learning, but also establishes a unified pipeline, achieving threat detection, automatic response, and explainability. The hierarchical design adopted not only reduces the reliance on professional technicians but also maintains modularity and scalability. The framework bridges the gap between the enterprise-level security technology requirements in a limited technical resource environment and actual deployment.

Our evaluation results indicate that our framework achieves a high detection accuracy (XGBoost accuracy 89%, AUC 0.98). At the same time, it significantly reduces the false alarm rate (5.4%). Moreover, it achieves an end-to-end fast response time of approximately 300 milliseconds. Although XGBoost achieves slightly higher accuracy (89%), Random Forest is selected for the operational workflow due to its lower computational overhead and faster inference, making it more suitable for real-time automation. A usability test for non-technical personnel indicates that non-technical personnel can effectively manage the security workflow with the support of the visualization dashboard and operation manual. This result verifies that the proposed framework has feasibility and practical value in popularizing AI-driven cloud security.

Contribution and scope. Instead of introducing a new standalone learning algorithm, this work focuses on system-level orchestration under real deployment constraints. (1) a reproducible closed-loop cloud-security orchestration pattern that couples SIEM ingestion, ML-based triage, and low-code

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response actions with an interpretable local dashboard for non-expert operators; (2) an empirically validated deployment-oriented model-selection rationale under resource constraints (accuracy–latency–maintenance trade-offs across XGBoost, Random Forest, and DNN); and (3) measurable operational improvements (end-to-end response latency and reduction of manual intervention) enabled by the proposed design. Our evaluation is conducted using the Microsoft GUIDE dataset augmented with controlled synthetic logs, and we therefore position robustness claims to the validated deployment setting while outlining additional cross-environment validation as future work. (4) We provide interpretable decision support for non-expert operators.

A. Reproducibility Information

A complete experimental resource package ensures the reproducibility of our research results. The source code and reproducibility package supporting the findings are openly available in Zenodo [18]: https://doi.org/10.5281/zenodo.17605067.

The full reproducibility package and deployment guide are publicly available at: https://github.com/weijunma/AI-Driven-Cloud-Security-Framework-SME-Healthcare. The package includes the following artifacts:

1) Experiment Manual and Setup Files

Step-by-step instructions, configuration scripts, and environment information for re-producing the experimental environment.

2) Source Code and Models

Python scripts for preprocessing, train-ing, and evaluation of the machine-learning models (including random forests, XGBoost, and DNN), together with the parameter settings used in our experiments.

3) Datasets

Subsets of the Microsoft Security Incident Prediction (GUIDE) dataset, along with synthetic attack logs used in our simulation study.

4) Dashboards

The localized Streamlit dashboard code and output pages of the running instance provide the following information: attack trends, event classification, and interpretability analysis.

5) Reproducibility Resources

A README file, deployment instructions, and configuration files that allow other researchers to re-run the experiments and validate the reported results.

B. Future Work

Although our results are exciting, our work has limitations. For instance, the reliance on controlled data sets is significant, and there is a lack of sensitivity to advanced threats. In addition, there is room for further optimization to improve usability during the deployment process. Future work should focus on verifying the framework in real-world environments. At the same time, based on the actual situation of the organization, the proposed architecture can be expanded to multi-cloud platforms such as AWS and GCP. Data sets from industries like finance and energy can also be considered. Finally, we will explore other methods to enhance the framework's adaptability and practicality to complex threats

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